

Assessment

Build and document a small project using any distributed system or cloud technology. The goal is hands-on exposure to real deployment/ architecture. Push your code to a public GitHub repo with a README explaining what you built, the architecture, and how to run it. Use GitHub Actions for CI/CD. Add basic test cases as well (at least 4) and block pushing of code if they fail.

The following are examples of projects: Pick ONE of these, or propose your own — as long as it involves: distributed communication, containers, cloud deployment, or message passing, etc.

1. **Containerized microservice on AWS** — Build a simple app, containerize it with Docker, push the image to ECR, and deploy it on an EC2 instance or ECS. README should include the architecture diagram (client → EC2/ECS → container) and steps to reproduce. (Theme: AWS/ deployment exposure specifically).
2. **Kafka-based event pipeline** — A simple producer-consumer setup: one script generates events (e.g., mock sensor data, order events, log lines) and publishes to a Kafka topic, another consumes and processes them (logging or writing to a file/DB). (Theme: message-queue/event-driven thinking, message delivery/ordering)
3. **Load-balanced multi-instance app** — Deploy the same simple app across 2+ EC2 instances (or containers) behind a load balancer, show requests being distributed. (Theme: visualising horizontal scaling, client-server paradigm)
4. **ML model deployment** — train a small model, wrap it in an API, containerize, deploy to EC2/ cloud. (Theme: client-server (API serving), containers, and cloud deployment)

Give a 2-3 minute presentation in class to explain the high-level working of your project like you would explain to an interviewer. Show your running deployed project.

Submission deadline for code: 30 September.

Submission link: https://docs.google.com/spreadsheets/d/158n95p7F09XiKvoSxIewYPQMSZ8hdg1U5dR02I_Bx3U/edit?usp=sharing